

PRODUCT REQUIREMENTS DOCUMENT

Agentic Fraud Intelligence & Analytics System

AFIAS | Enterprise AI Fraud Intelligence Platform | Version 1.0

Document Version	Classification	Status	Date
1.0 – Initial Release	CONFIDENTIAL	APPROVED	April 2026
Product Owner	Architecture Lead	Security Lead	AI/ML Lead
Enterprise Product	Platform Engineering	Zero-Trust IAM	AI Agents Team

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1. Executive Summary

The Agentic Fraud Intelligence & Analytics System (AFIAS) represents the apex of Enterprise Digital Transformation — merging a unified data foundation with a Zero-Trust security perimeter and frontier Agentic AI to deliver end-to-end autonomous fraud investigation.

AFIAS transforms the Snowflake AI Semantic Layer into a governed gateway for autonomous Investigator Agents, orchestrating a multi-model reasoning system that replaced manual, rule-based detection with an intelligent, self-correcting engine:

- Claude 3.5 (AWS Bedrock) — strategic fraud hypothesis reasoning and cross-domain pattern synthesis
- Gemini 1.5 Pro (Google Vertex AI) — deep-log behavioral analysis and temporal anomaly detection
- GPT-4o (LLM-as-Judge) — automated quality validation ensuring every autonomous decision is grounded in Enterprise Golden Truth

The entire system operates within a Zero-Trust IAM architecture enforcing RBAC/ABAC policies at every boundary. The outcome is transformational: the fraud investigation lifecycle has been compressed from a 72-hour manual process to a sub-10-minute autonomous investigation, directly protecting the product lifecycle and accelerating business growth through scalable, verified intelligence.

1.1 Business Impact Snapshot

KPI	Before AFIAS	After AFIAS (Target)
Fraud Investigation Cycle	72 hours (manual)	< 10 minutes (autonomous)
False Positive Rate	~35%	< 8% (LLM-as-Judge validated)
Analyst Case Load	100% manual review	< 15% exception-only review
Decision Auditability	None / ad hoc	100% traced & logged
System Availability	Business hours only	99.95% uptime (24 x 7)
Cross-Model Consensus Score	N/A	>= 0.85 confidence threshold

2. Problem Statement

Enterprise fraud operations face a structural crisis driven by three converging forces:

2.1 Scale & Velocity Mismatch

Modern fraud patterns evolve faster than human analysts can respond. Transaction volumes have scaled exponentially while adversarial tactics — synthetic identity fraud, account takeover, first-party fraud collusion — have become more sophisticated and coordinated. A 72-hour manual investigation cycle creates an unacceptable window of exposure.

2.2 Data Fragmentation

Fraud signals are distributed across siloed data systems: transaction platforms, identity stores, behavioral event streams, device telemetry, and third-party risk signals. Without a unified semantic layer, investigators spend the majority of their time in data assembly rather than analysis. This creates inconsistency, blind spots, and non-reproducible decisions.

2.3 Rule-Based Detection Decay

Legacy rule-based fraud detection systems are inherently reactive and brittle. Each new fraud vector requires manual rule authoring, testing, and deployment — a cycle that consistently lags attacker innovation. False positive rates approaching 35% result in investigator fatigue, customer friction, and eroded trust in automated systems.

2.4 Governance & Auditability Deficit

Regulatory and internal audit requirements demand explainable, reproducible fraud decisions. Manual processes produce inconsistent documentation. Existing ML models operate as black boxes with no native auditability. The absence of an immutable decision trail creates material compliance and litigation risk.

3. Goals & Non-Goals

3.1 Goals

- Deliver autonomous, end-to-end fraud investigation with sub-10-minute cycle time from signal to disposition
- Unify all fraud-relevant data through the Snowflake AI Semantic Layer as the Enterprise Golden Truth
- Orchestrate a multi-model AI reasoning system (Claude 3.5, Gemini 1.5 Pro) for comprehensive fraud analysis
- Implement GPT-4o LLM-as-Judge quality validation ensuring decision accuracy and hallucination prevention
- Enforce Zero-Trust IAM (RBAC/ABAC) across every agent action, data access, and model invocation
- Produce immutable, audit-ready decision logs satisfying regulatory compliance requirements
- Reduce analyst workload through intelligent escalation — humans review exceptions, not every case
- Deliver real-time executive intelligence on fraud trends, system performance, and business impact

3.2 Non-Goals

- AFIAS is not a payment processing or transaction authorization system
- AFIAS does not replace human judgment in high-stakes edge cases — it augments and escalates
- AFIAS does not modify production data or execute account actions autonomously (read + recommend only)
- Real-time streaming at sub-second latency is not a current-phase objective; batch-triggered near-real-time is in scope
- Consumer-facing fraud prevention UI (chargebacks, dispute portals) is out of scope for this PRD

4. User Personas & Value Proposition

Persona	Core Need	AFIAS Value Delivered
Fraud Investigator	Rapid, evidence-backed case decisions	Pre-investigated case packets with confidence scores in <10 min
Risk & Compliance Officer	Provable, audit-ready decision trails	Immutable logs, GPT-4o judge attestation, regulatory export
Data/ML Engineer	Governed data access for AI workloads	Snowflake Semantic Layer as single source of truth
CISO / Security Architect	Zero-Trust enforcement across AI agents	RBAC/ABAC policy engine gating every inference call
Executive Leadership	Measurable fraud loss reduction & ROI	Real-time executive dashboards, fraud loss trending, autonomous throughput

5. System Architecture

AFIAS is a layered, event-driven architecture comprising seven tightly integrated components. Every component operates within the Zero-Trust security perimeter and contributes to an immutable audit trail.

5.0 Architecture Overview Diagram

The diagram below shows the complete AFIAS system architecture: data ingestion sources converging into the Snowflake AI Semantic Layer, the multi-agent orchestration tier, the GPT-4o quality gate, and the output and audit layer — all enclosed within the Zero-Trust IAM perimeter.

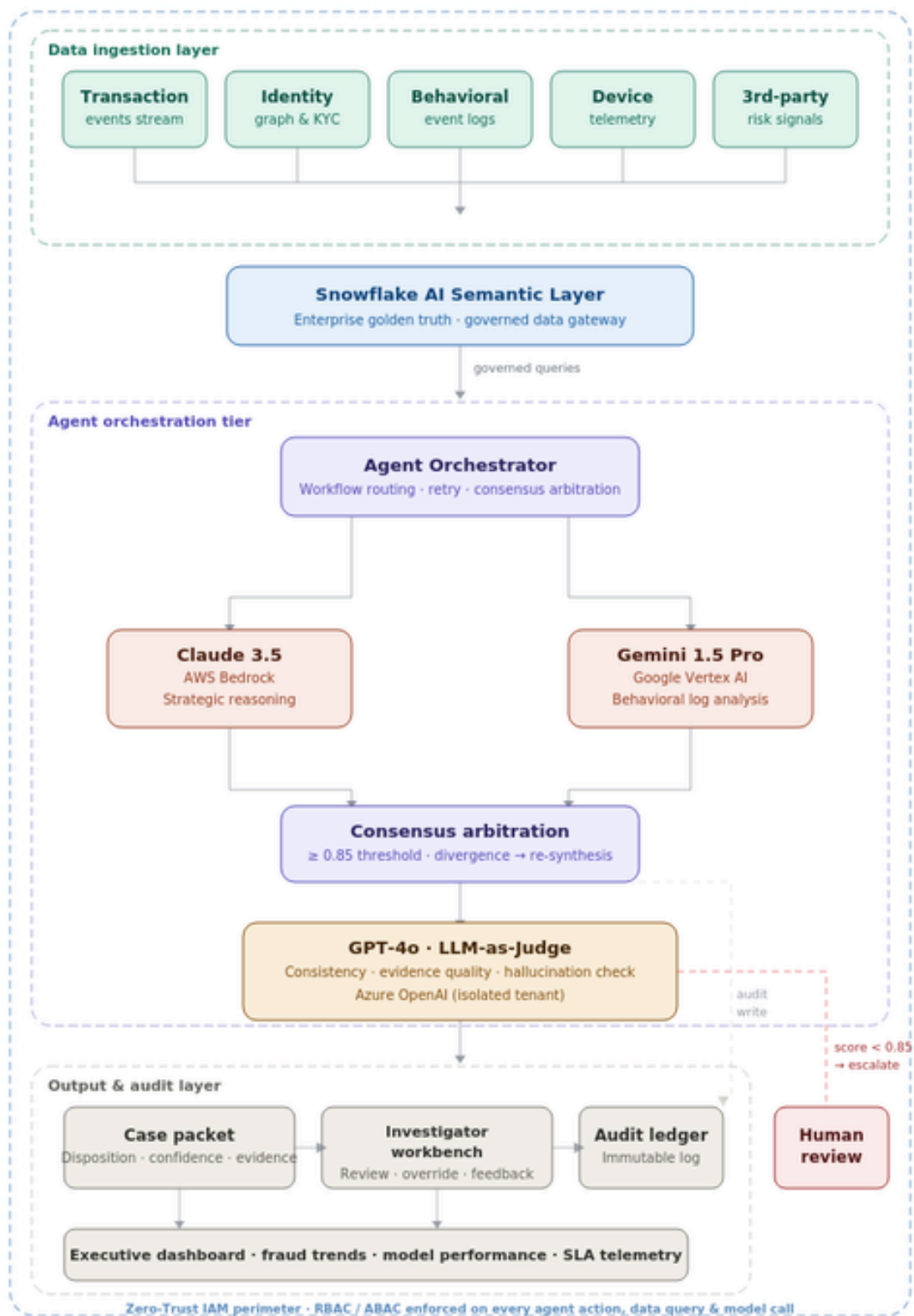


Figure 5.1 — AFIAS System Architecture: Data Ingestion, Agent Orchestration & Audit Pipeline

Green — Data ingestion	Purple — Agent orchestration	Amber — Quality gate	Gray — Output & audit
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5.1 Architecture Component Reference

Layer	Technology	Responsibility
Data Foundation	Snowflake AI Semantic Layer	Unified governed data gateway; single source of enterprise truth for all agent queries
Strategic Reasoning	Claude 3.5 (AWS Bedrock)	High-level fraud hypothesis generation, cross-domain pattern synthesis, case summarization
Behavioral Analysis	Gemini 1.5 Pro (Google Vertex AI)	Deep log analysis, sequential transaction profiling, temporal anomaly detection
Quality Assurance	GPT-4o (LLM-as-Judge)	Automated validation of agent outputs; scores confidence, flags hallucinations, triggers re-investigation
Security Perimeter	Zero-Trust IAM (RBAC/ABAC)	Identity-first policy enforcement on every agent action, data query, and model call
Orchestration	Agent Orchestrator Service	Workflow coordination, model routing, retry logic, consensus arbitration between models
Observability	Audit Ledger + Analytics	Immutable decision audit trail, real-time dashboards, SLA monitoring, compliance export

5.2 Data Flow — Autonomous Investigation Pipeline

The AFIAS investigation pipeline executes the following sequence upon fraud signal detection:

- Step 1 — Signal Ingestion: Fraud event enters the pipeline via Snowflake Semantic Layer data gateway. RBAC/ABAC policy engine validates the agent's access entitlement.
- Step 2 — Context Assembly: Orchestrator queries the Semantic Layer for transaction history, identity graph, behavioral baseline, and peer cohort benchmarks.
- Step 3 — Strategic Reasoning: Claude 3.5 (Bedrock) receives the assembled context and generates a structured fraud hypothesis with supporting evidence citations.

- Step 4 — Behavioral Deep-Dive: Gemini 1.5 Pro (Vertex AI) performs sequential log analysis, identifying temporal anomalies and micro-behavioral deviations across the transaction chain.
- Step 5 — Consensus Arbitration: The Orchestrator compares Claude and Gemini outputs. Where consensus exceeds 0.85, a preliminary disposition is formed. Divergence triggers a third-pass synthesis.
- Step 6 — Quality Gate: GPT-4o LLM-as-Judge evaluates the combined output for logical consistency, evidence quality, and hallucination risk. Outputs below threshold are routed to human escalation.
- Step 7 — Case Packet Generation: A structured case packet is produced containing disposition, confidence score, evidence chain, model reasoning traces, and recommended action.
- Step 8 — Audit Ledger Write: Every step — inputs, outputs, model versions, timestamps, policy evaluations — is written to the immutable audit ledger.
- Step 9 — Investigator Delivery: The case packet surfaces in the Investigator Workbench. High-confidence cases auto-close. Escalated cases queue for human review.

5.3 Zero-Trust IAM Architecture

All AFIAS components operate on a Zero-Trust principle: no implicit trust, continuous verification, least-privilege access. The IAM layer enforces:

- RBAC (Role-Based Access Control): Agents, services, and human users are assigned minimum-necessary roles. Agent roles are scoped to specific data domains and model endpoints.
- ABAC (Attribute-Based Access Control): Dynamic policy evaluation factors in context attributes — case sensitivity classification, data jurisdiction, model confidence level, and time-of-day — to govern access decisions.
- Service-to-Service mTLS: All inter-service calls are mutually authenticated with rotating certificates. No service trusts network-layer identity alone.
- Policy-as-Code: All IAM policies are version-controlled, tested in CI/CD, and deployed through automated pipelines — eliminating manual configuration drift.

6. Product Requirements

6.1 Functional Requirements

ID	Priority	Requirement	Owner
FR-01	P0	Ingest and normalize transaction, identity, and behavioral event streams from Snowflake Semantic Layer in real time	Data Engineering
FR-02	P0	Trigger autonomous investigation pipeline upon fraud signal detection within 60 seconds of event ingestion	Agent Platform
FR-03	P0	Invoke Claude 3.5 (AWS Bedrock) for strategic fraud hypothesis reasoning per investigation	AI/ML
FR-04	P0	Invoke Gemini 1.5 Pro (Google Vertex AI) for deep behavioral log analysis per investigation	AI/ML
FR-05	P0	Execute GPT-4o LLM-as-Judge validation on every agent output prior to case disposition	QA Platform
FR-06	P0	Enforce Zero-Trust RBAC/ABAC policies on every data access, model invocation, and agent action	Security
FR-07	P1	Generate structured case packets with evidence citations, confidence scores, and recommended disposition	Agent Platform
FR-08	P1	Present Investigator Workbench UI enabling human review, override, and feedback capture	Product
FR-09	P1	Produce immutable, timestamped audit log for every autonomous decision and model output	Compliance
FR-10	P1	Provide real-time executive dashboard with fraud loss trends, throughput, and model performance	Analytics
FR-11	P2	Support configurable confidence thresholds to escalate	Agent Platform

		borderline cases to human review	
FR-12	P2	Enable bulk case export in regulatory-compliant formats (CSV, JSON, PDF) for audit purposes	Compliance

6.2 Non-Functional Requirements

ID	Priority	Requirement	Owner
NF-01	P0	End-to-end investigation latency < 10 minutes from signal detection to case disposition	Platform
NF-02	P0	System availability 99.95% uptime with < 5-minute MTTR for critical pipeline components	Infrastructure
NF-03	P0	All inter-service communication encrypted (TLS 1.3+); all data encrypted at rest (AES-256)	Security
NF-04	P0	Zero-Trust policies enforced with sub-100ms latency overhead per policy evaluation	Security
NF-05	P1	Horizontal scalability to process 10,000 concurrent investigations without degradation	Platform
NF-06	P1	LLM-as-Judge quality gate achieving >= 0.85 consensus confidence score threshold	AI/ML
NF-07	P1	Audit logs immutable, tamper-evident, and retained for minimum 7 years	Compliance
NF-08	P2	Model response cache hit rate >= 60% for repeated behavioral pattern queries	AI/ML

7. AI Model Specification

7.1 Claude 3.5 — Strategic Fraud Reasoning Agent

Platform: AWS Bedrock

Claude 3.5 serves as the Strategic Reasoning Agent, responsible for high-order fraud hypothesis formation. Its responsibilities include:

- Synthesizing multi-domain context (transaction, identity, behavioral, network) into a coherent fraud narrative
- Generating ranked fraud hypotheses with supporting evidence citations drawn exclusively from Snowflake Semantic Layer data
- Identifying cross-case fraud ring patterns through entity relationship reasoning
- Producing structured outputs conforming to the AFIAS Case Schema (JSON) for downstream orchestration

Guardrails: Claude is constrained by a system prompt that enforces evidence-grounded reasoning, prohibits speculation beyond provided data, and mandates uncertainty acknowledgment when signal is insufficient.

7.2 Gemini 1.5 Pro — Behavioral Log Analysis Agent

Platform: Google Vertex AI

Gemini 1.5 Pro serves as the Behavioral Analysis Agent, leveraging its extended context window for deep sequential analysis:

- Processing complete transaction log sequences (up to 1M token context) without truncation
- Detecting micro-behavioral deviations: velocity anomalies, session fingerprint drift, device characteristic changes
- Building behavioral baseline profiles and scoring deviation magnitude against peer cohort distributions
- Identifying temporal attack chains — sequences of actions that individually appear benign but collectively signal fraud

7.3 GPT-4o — LLM-as-Judge Quality Assurance

Platform: Azure OpenAI (isolated tenant)

GPT-4o operates as an independent validation layer, not a fraud reasoning agent. Its responsibilities are strictly evaluative:

- Scoring multi-model reasoning outputs across four dimensions: logical consistency, evidence sufficiency, confidence calibration, and hallucination risk
- Computing a composite Quality Score (0.0 – 1.0); outputs below 0.85 threshold trigger human escalation
- Identifying disagreement vectors between Claude and Gemini outputs and classifying their nature (data gap, reasoning divergence, ambiguous evidence)
- Generating an auditable Judge Attestation record that travels with every case disposition

Independence principle: GPT-4o receives only the structured agent outputs — not raw data or system prompts — ensuring its evaluation remains free from anchoring bias.

8. Security & Compliance

8.1 Zero-Trust Security Principles

- Verify explicitly: Every request authenticated and authorized regardless of network location
- Use least privilege: Agent permissions scoped to minimum required for specific investigation task
- Assume breach: All traffic treated as potentially hostile; lateral movement impossible by design
- Continuous validation: Session tokens re-validated at configurable intervals; no persistent trust

8.2 Data Security

- All data in transit encrypted with TLS 1.3+; all data at rest encrypted with AES-256
- PII data anonymized or tokenized before passing to external model APIs (Bedrock, Vertex AI, Azure OpenAI)
- Snowflake Semantic Layer enforces column-level security; agents cannot access fields outside their permission scope
- Data residency policies enforced by ABAC — jurisdictionally restricted data never leaves compliant boundaries

8.3 Regulatory Alignment

- SOC 2 Type II — controls mapped to availability, confidentiality, processing integrity, and privacy
- GDPR / CCPA — data minimization, right-to-erasure workflows, lawful basis documentation
- PCI-DSS — cardholder data handling governed by dedicated agent role with elevated audit requirements
- Financial regulatory bodies — model explainability requirements met through LLM-as-Judge attestation chain

9. Risks & Mitigations

ID	Risk	Severity	Mitigation
R-01	LLM hallucination leading to incorrect fraud disposition	HIGH	GPT-4o LLM-as-Judge gate; human escalation for <0.85 confidence; full evidence citation required
R-02	Model vendor API downtime (Bedrock/Vertex AI)	HIGH	Circuit-breaker pattern with automatic failover; degraded mode with single-model operation
R-03	Snowflake Semantic Layer data drift or schema change	MEDIUM	Automated schema validation on ingestion; data contract testing in CI/CD pipeline
R-04	Zero-Trust policy misconfiguration blocking legitimate agent access	MEDIUM	Policy-as-code with automated regression tests; staged rollout with shadow mode validation
R-05	Regulatory change invalidating autonomous decision authority	MEDIUM	Configurable human-in-the-loop threshold; full auditability by design; legal review cadence
R-06	Adversarial fraud patterns evading multi-model detection	MEDIUM	Continuous model retraining pipeline; red-team adversarial testing quarterly

10. Delivery Roadmap

AFIAS is delivered across six sequential phases over a 9-month programme. Each phase has defined exit criteria that must be met before the next phase commences.

Phase	Milestone	Timeline	Exit Criteria
1	Foundation – Snowflake Semantic Layer + Zero-Trust IAM deployment	Months 1–2	Data gateway live; RBAC/ABAC policies enforced; >95% data quality score
2	Agent Core – Claude 3.5 & Gemini 1.5 Pro orchestration pipeline	Months 3–4	Dual-model pipeline processes 100 test cases; latency <10 min; outputs structured
3	Quality Gate – GPT-4o LLM-as-Judge integration & calibration	Month 5	Judge consensus ≥ 0.85 on 500-case validation set; false positive delta documented
4	Workbench – Investigator UI & Audit Ledger	Month 6	Investigators complete UAT; audit log immutability verified; export functions validated
5	Production Hardening & Compliance Sign-Off	Months 7–8	99.95% uptime SLA met in 30-day load test; regulatory sign-off obtained
6	GA Launch & Executive Dashboard	Month 9	All KPIs baselined; executive dashboards live; incident response runbooks published

11. Success Metrics & Evaluation

11.1 Operational Metrics

- P50 / P95 investigation cycle time (target: P50 <5 min, P95 <10 min)
- Autonomous disposition rate (target: >85% cases closed without human intervention)
- False positive rate (target: <8%; measured by Investigator feedback on auto-closed cases)
- LLM-as-Judge quality score distribution (target: median >0.90; zero dispositions below 0.70)
- System availability and pipeline SLA breach rate

11.2 Business Metrics

- Fraud loss prevented (dollar value) compared to pre-AFIAS baseline
- Analyst hours recaptured per quarter (time previously spent on manual investigation)
- Customer friction reduction (false-positive-driven account blocks down)
- Regulatory audit findings (target: zero material findings related to decision auditability)

11.3 Model Performance Metrics

- Cross-model consensus rate (Claude vs Gemini agreement on disposition direction)
- Model drift detection — weekly automated evaluation against a held-out labeled dataset
- Adversarial robustness score — quarterly red-team evaluation measuring novel fraud pattern detection rates

12. Appendix

12.1 Glossary

- AFIAS: Agentic Fraud Intelligence & Analytics System
- RBAC: Role-Based Access Control — permissions granted based on assigned role
- ABAC: Attribute-Based Access Control — permissions evaluated dynamically against contextual attributes
- LLM-as-Judge: Pattern of using a large language model to evaluate the output quality of other LLMs
- Semantic Layer: An abstraction layer over raw data that applies business definitions, governance, and access controls
- Zero-Trust: Security model requiring continuous verification of every access request, regardless of network location
- Enterprise Golden Truth: The authoritative, governed version of enterprise data as surfaced through the Snowflake Semantic Layer
- Consensus Score: A composite metric (0.0–1.0) reflecting agreement between Claude and Gemini outputs on fraud disposition

12.2 Related Documents

- AFIAS Technical Architecture Design Document (TDD-AFIAS-001)
- Zero-Trust IAM Policy Specification (SEC-IAM-AFIAS-001)
- Snowflake AI Semantic Layer Data Contract (DATA-CONTRACT-001)
- LLM-as-Judge Calibration Report (AI-QA-JUDGE-001)
- AFIAS Security Threat Model (SEC-THREAT-001)
- Regulatory Compliance Mapping Matrix (COMPLIANCE-MAP-001)

END OF DOCUMENT — AFIAS PRD v1.0

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